

Renhao (Lukas) Xue

ML / Research Engineer · LLM Post-Training & Inference

Seattle, WA · lukas.rxue@gmail.com · 470-775-1452 · lukasxue.com ·   

EXPERIENCE

Amazon Web Services, Generative AI Innovation Center · Seattle, WA

Machine Learning Engineer II, Custom Model Optimization · May 2025 – Present

- Post-trained foundation models **up to 30B parameters** (Qwen3, Llama 3.1, Nova 2 Lite; text and vision-language) with SFT, DPO, and GRPO — from single-GPU LoRA (Unsloth) to distributed SageMaker HyperPod runs on 8×H200 and 8×A100 clusters — shipping custom models to enterprise production.
- Fine-tuned **Nova 2 Lite** (VLM) for document understanding (classification, key-information extraction, packet splitting) via single-phase shuffled-heavy fine-tuning: **matched Claude Sonnet 4.5 quality at ~1/10 the inference cost**, with 2× ordered-split accuracy and 96% page classification on shuffled multi-page PDFs.
- Built distributed multi-node/multi-GPU training and inference for a single-cell foundation model at a premier research institute (**77–90% training speedups**); a RAPIDS cuML + Dask pipeline cut cell-type evaluation from hours to minutes; reference architecture adopted across teams.
- **3.7× peak throughput** (70 → 261 req/s) and **39% lower P50 latency** for the Large Payment Model, a top-10 global bank's fraud-detection foundation model, vs the bank's FP32 PyTorch baseline: TensorRT on Triton with dynamic batching, lifting GPU utilization from ~40% to 90%+; INT8 QAT added +68% throughput at near-lossless embedding fidelity.
- Profiled with Nsight Systems to isolate CPU-side bottlenecks, then vectorized concurrent tokenization and bucketized CUDA Graphs to cut kernel-launch overhead; designed the burst-absorption serving path (~370 requests / 50 ms: bounded per-pod batching queues with backpressure, warm pods, EKS auto-scaling). **In production, accelerating go-live by 3 months.**
- Built a text-to-PySpark agentic workflow for an enterprise networking data lake: DSPy auto-prompt optimization (**+46% CodeBLEU**), complexity-aware query routing (+43% on complex queries), and LangGraph workflow optimization (–19% end-to-end latency).

Software Development Engineer, Amazon Bedrock / SageMaker · promoted to SDE II · Feb 2023 – May 2025 (Intern, 2022)

- **Led and shipped on-demand, token-based serving of fine-tuned models to production:** multi-tenant LoRA serving (shared base model in memory, per-customer adapters hot-swapped at runtime) with KV-cache-aware request routing — **92–99% lower inference cost** for spiky low-traffic workloads and 90% faster deployment than provisioned throughput, driving adoption of Bedrock model customization.
- Shipped **Llama 3 SFT and distillation** in Bedrock Model Customization (**launched at re:Invent 2024**); cut fine-tuning job failures **50%** with an API-layer validation system.

PUBLICATIONS

- C. Ling, et al. (incl. **R. Xue**). *Deep Graph Representation Learning and Optimization for Influence Maximization*. **ICML 2023**. [arXiv](#)
- R. Wang*, **R. Xue*** et al. *PRISM: Position-encoded Regressive Inverse Spectral Model for Multilayer Thin-Film Design*. ICML Workshop on AI for Physics, 2026. (*equal contribution) [arXiv](#)
- R. Wang, **R. Xue**, et al. *AME-TS: Anchored Mixture-of-Experts for Time Series Forecasting*. ICML 2026 Workshop; extended version under review at NeurIPS. [arXiv](#)
- Y. Cui, **R. Xue**, et al. *From Errors to Rules: Iterative Prompt Optimization for Text Classification*. Preprint, 2026. [arXiv](#)
- **R. Xue** et al. *Inference-Time Model Steering via Predictive-State Intervention: A Survey*. Preprint, 2026. [preprint](#)

OPEN SOURCE

- [awslabs/mcp](#): Code owner and recurring contributor to AWS's official Model Context Protocol repo.
- [nanoLLaDA](#): Minimal (~500-line) masked diffusion language model · [PRISM](#): inverse thin-film design transformer (interactive playground) · [awesome-inference-time-steering](#): curated survey.

EDUCATION

- **B.S. Computer Science** (Minor: Applied Math), **Emory University** · 2019 – Dec 2022 · GPA 3.94/4.00. Student researcher (graph neural networks); TA for ML (CS334) and Algorithms (CS326).
- M.S. Information Studies, Trine University (part-time) · 2025 – 2026

TECHNICAL SKILLS

Post-Training: SFT, DPO, GRPO, CPT, distillation, LoRA/adapters, quantization (QAT/PTQ)

Frameworks: PyTorch, HuggingFace (Transformers/PEFT/TEI), Lightning, Unsloth, FSDP, DDP, Dask, RAPIDS, DSPy, LangGraph

Inference / Serving: vLLM, TensorRT, Triton, ONNX Runtime, CUDA graphs, dynamic batching · **Distributed:** multi-node/multi-GPU, data/model parallelism, SageMaker HyperPod

Infra / Profiling: AWS (Bedrock, SageMaker, EKS, MSK, Lambda, FSx), Kubernetes, Docker, Terraform, Nsight, DCGM, Prometheus, Grafana, OpenTelemetry · **Languages:** Python, Java, SQL, TypeScript